

Start-up Financing, Entry and Innovation

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DSE 2026: Dynamic Finance and Banking

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and then use that to solve everything else.*

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— DeepMind

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 - Do high failure rates and cross-country differences reflect **investment opportunities** or the **allocation of funding**?
- **Challenge.** Distinguishing fundamentals from finance.
- **Approach.** An **equilibrium model** with matching and staged financing.
 - Staging limits losses when entrepreneurs **hide failure**...
 - ...but exposes firms to **financing risk**, which can induce premature closure.

Headline results

- **Measurement.** Funding histories reveal project characteristics and financing conditions.

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- **US decomposition.** $\sim 40\%$ **shut down prematurely.**
- **UK vs. US comparison.**
 - **Financing** $\rightarrow$ $2/3^{\text{rd}}$ **entry gap**; acquisitions important, but **not project quality**.
 - **Policy.** Reallocating capital to later-stage start-ups improves outcomes.

VC in entrepreneurial finance. Sahlman (1990), Admati and Pfleiderer (1994), **Gompers (1995)**, Bergemann & Hege (1998), Kaplan & Strömberg (2003), Sørensen (2007), Tian (2011), **Nanda & Rhodes-Kropf (2013, 2017)**, Ewens et al. (2022).

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- **Contribution.** “Operational” theory: identification and counterfactuals.

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Equilibrium models.

- **Theoretical.** Inderst & Müller (2004), Michelacci & Suarez (2004), Jovanovic & Szentes (2013), Hellmann & Thiele (2015), Sannino (2024).
- **Quantitative.** Ates (2018), Akcigit et al. (2022), Greenwood et al. (2022), Ando (2024).

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Economic + financial development. King & Levine (1993), Rajan & Zingales (1998), Buera et al. (2011), Moll (2014), Cavalcanti et al. (2023), **Aragoneses and Saxena (2025).**

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Equilibrium models.

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- Quantitative.
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Economic + financial development.

- **Contribution.** Setting where self-financing from retained earnings is infeasible.

Tractable equilibrium model

- Difficult to evaluate projects

Matching friction

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- VCs stage capital injections

Private benefits and information → hidden failure

Staging is the **most important** control mechanism
(Sahlman, 1990; Hall & Lerner, 2010).

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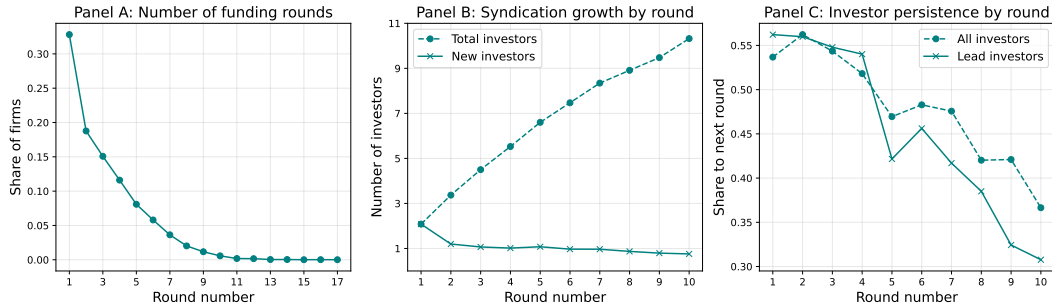
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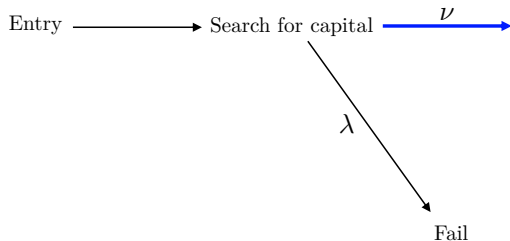
“Raise as much as possible.”
Sequoia, RIP Good Times (2008)

Staging and investor syndication



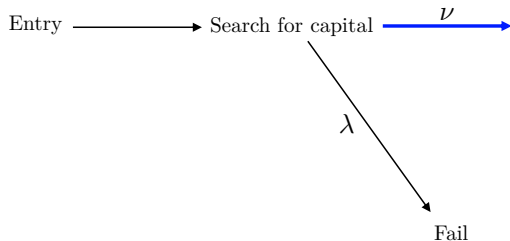
Source: Thomson Reuters, US VC-backed start-ups first funded 2005–2015, data to end-2022. Censored at 7 years (panel A) and 12 years (panel B).

Frictions in securing external funding



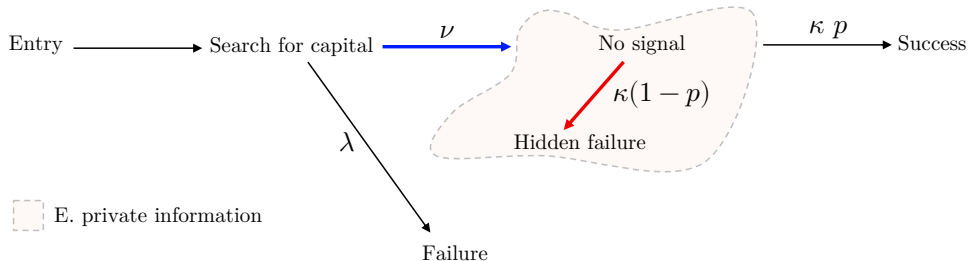
- Entrepreneur must **match with a VC**
 - Endogenous **meeting rate**, ν .
 - Without funding, **shuts down** at rate λ .

Frictions in securing external funding



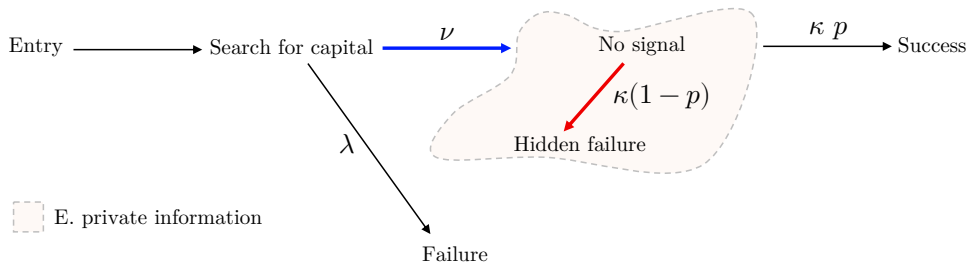
- Upon matching, **project characteristics** (k, κ, p, π) are common knowledge.
 - Flow investment, k .
 - Information arrives at rate κ , date $T_\kappa \sim \text{Exp}(\kappa)$.
 - Success with probability p and payoff π ; otherwise zero.

Frictions in securing external funding



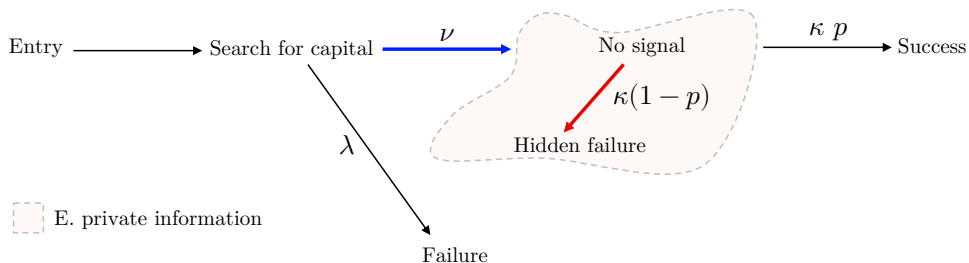
- Entrepreneur **privately observes R&D signals** and derives **private benefits**.
 - **Hidden failure**: private information + benefits $\rightarrow$ incentive to conceal bad news.

Frictions in securing external funding



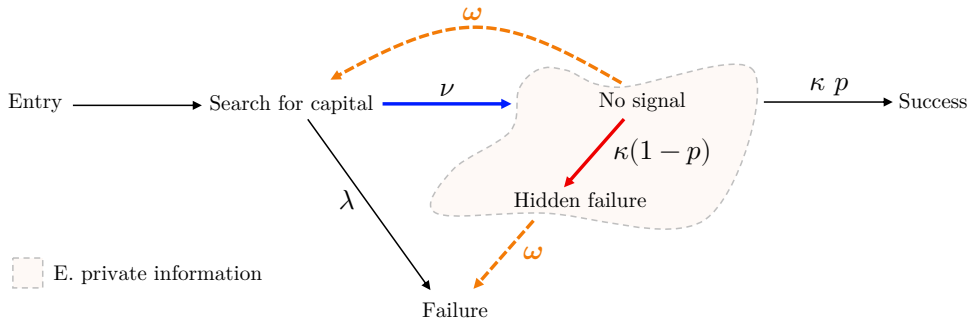
- **Equity contract** (ς, ω) . VC gets share ς ; funding commitment **ends at rate ω** .
 - Expected **runway** is $1/\omega$.

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Solving for the contract

- Gross (post-money) value, $V_d(\omega)$, of a funded start-up:

$$\rho V_d(\omega) = \kappa[p\pi - V_d(\omega)] + \omega[V_s - V_d(\omega)].$$

where $V_s \leq V_d(\omega)$ is the value in search.

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Shorter runway ($\omega \uparrow$) raises exposure to **financing risk**.

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$$K(\omega) = \frac{k}{\rho + \kappa + \omega} \left(1 + \frac{\kappa(1-p)}{\rho + \omega} \right)$$



Shorter runway ($\omega \uparrow$) reduces
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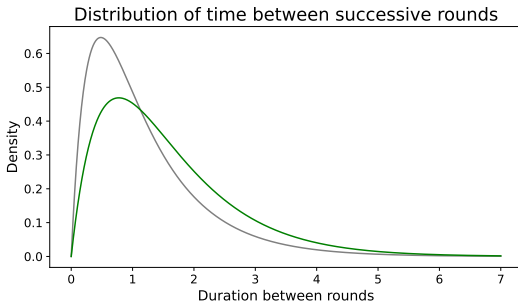
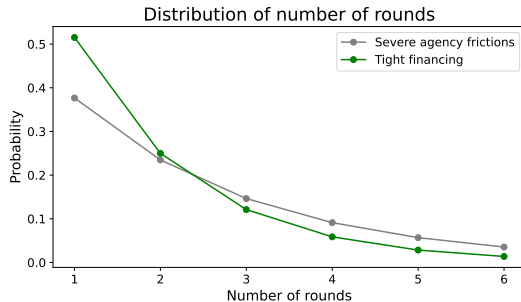
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- **Contracting problem.** The entrepreneur chooses the contract rate, ω :

$$V^M = \sup_{\omega \in [0, \infty)} \{ (1 - \varsigma) V_d(\omega) \}, \quad \text{s.t.} \quad \underbrace{\varsigma V_d(\omega) = K(\omega)}_{\text{VC participation constraint}}$$

Equilibrium characterisation

Funding histories



- **Number of rounds**, $N_f \sim \text{Geometric}$ with $\frac{\partial \mathbb{E}[N_f]}{\partial \omega} > 0$.
- **Duration between rounds**, $T_{br} \sim \text{Hypoexponential}$ with $\frac{\partial \mathbb{E}[T_{br}]}{\partial \omega} < 0$.

Measurement

An identification result

- Consider the following set of moments, $\mathcal{M}$
 - Funding histories: $\mathbb{E}[N_f], \mathbb{E}[T_{br}]$.
 - Outcomes: $\mathbb{E}[1\{\text{success}\}], \mathbb{E}[\text{time-to-exit}], \mathbb{E}[\text{exit multiple}]$.
 - Investment: $\mathbb{E}[\text{burn rate}]$.

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Proposition 1

Given $\mathcal{M}$, the structural parameters $\theta = (\kappa, p, \pi, k, \nu, \lambda)$ can be recovered uniquely.

Quantitative model

To a quantitative model

- **Minimal extensions**
 - Early- and late-stage development; acquisitions as an alternative exit.

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- Minimal extensions
 - Early- and late-stage development; acquisitions as an alternative exit.
- Data
 - Thomson Reuters funding and exit histories.
 - US firms first funded 2005–2015; outcomes through 2022; **11,022 firms**.

To a quantitative model

- **Minimal extensions**

- Early- and late-stage development; acquisitions as an alternative exit.

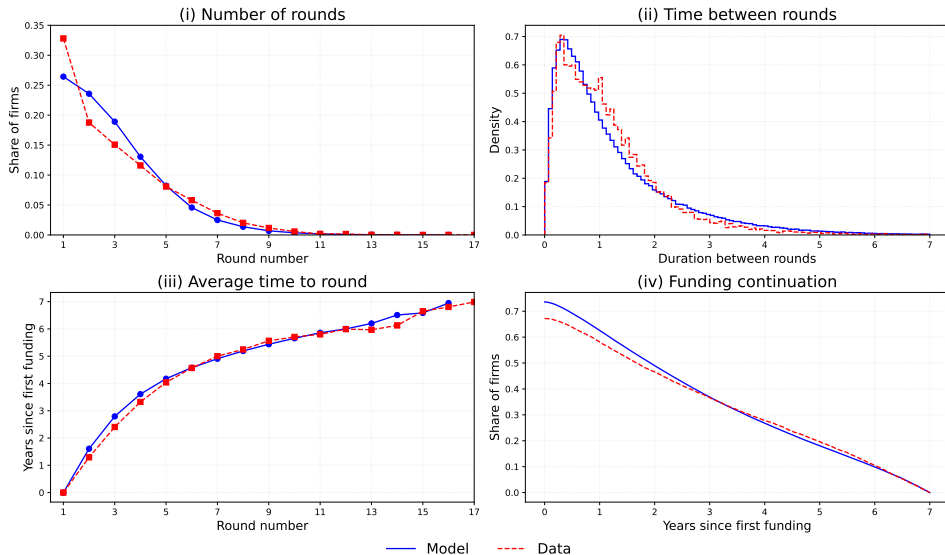
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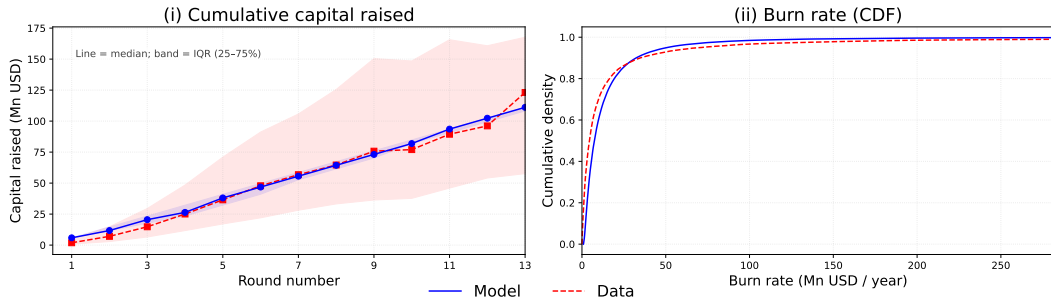
- **Estimation**

- Simulated method of moments using funding histories, capital use and exits.

Assessing model fit: funding histories

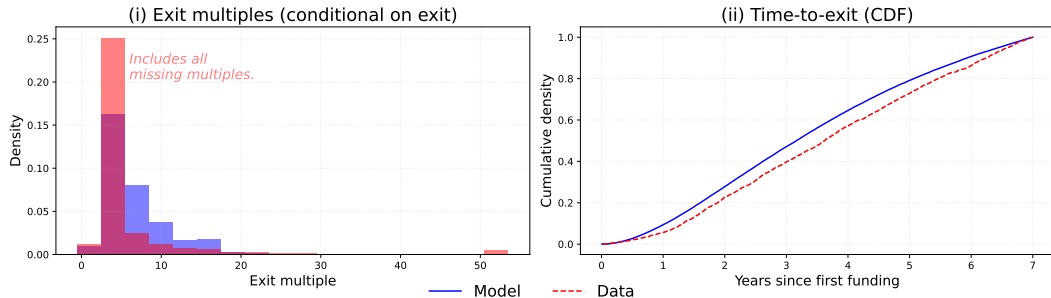


Assessing model fit: capital intensity



Notes: The left-hand panel reports cumulative capital raised by funding round. The right-hand panel reports the distribution of burn rates. Data are censored seven years after first funding.

Assessing model fit: outcomes



Notes: The left-hand side reports the exit-multiple distribution, conditional on exit. The right-hand side reports the time-to-exit CDF. All data are censored at seven years following first funding.

Two takeaways

1. Substantial *ex ante* uncertainty

- Model features *ex ante* homogenous start-ups, yet captures rich *ex post* heterogeneity in outcomes.
- Structural counterpart to evidence of weak *ex ante* predictability (given investment).
 - Kerr, Nanda, and Rhodes-Kropf (2014); Kerr, Lerner, and Schoar (2014).

Two takeaways

2. Financing-driven failures are significant

- Baseline outcomes: **70.6% of firms fail**; 4.9% have “success”; 24.5% are acquired.
 - **Premature termination rate: 40.6%**. Majority in early-stage.

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“The second biggest cause of death is probably the difficulty of raising money. Fundraising is brutal.” — Paul Graham (Y Combinator)

Two takeaways

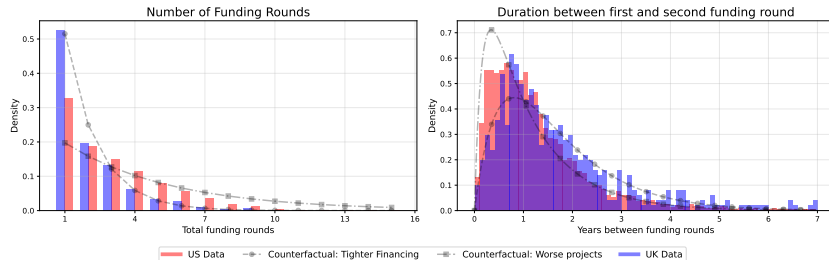
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 - **Premature termination rate: 40.6%**. Majority in early-stage.
- Removing financing-driven failure (frictionless benchmark)
 - Success rate: **5pp to 11pp** (+6pp).
 - Acquisition rate: **25pp to 39pp** (+14pp).

Venture capital across countries

UK start-ups face tighter financing and acquisition markets

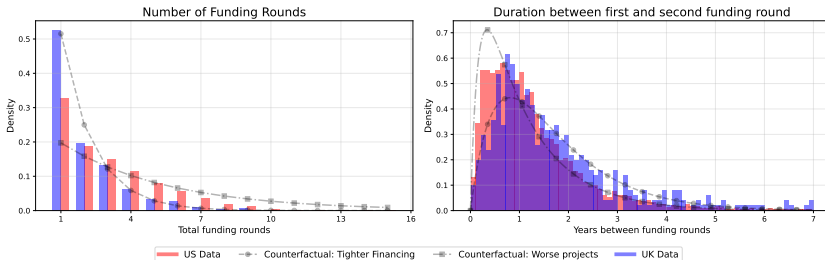
An intuitive explanation



- UK start-ups raise **fewer rounds**: 2.1 vs. 3.0 within seven years.
- UK start-ups go **longer between rounds**: 17 months vs. 14 months.

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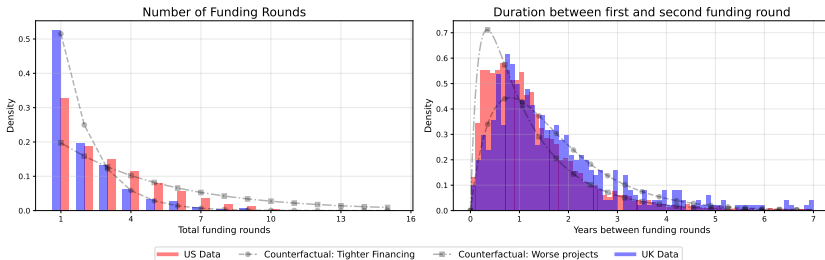
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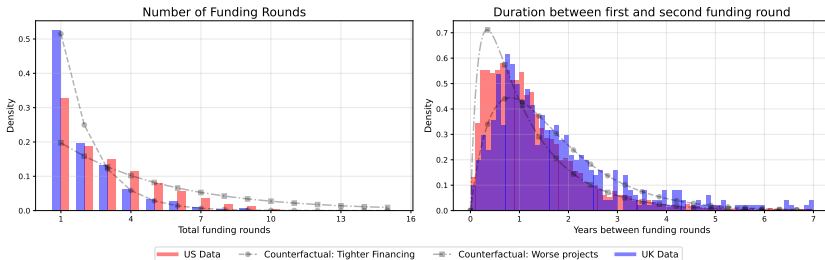
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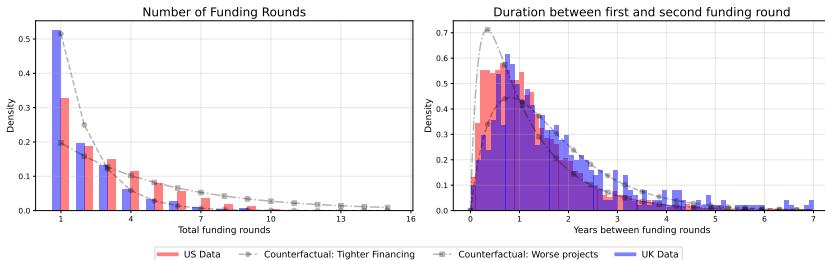
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Tight financing + lower acquisition rate $\Rightarrow$ **thin acquisition market**

US versus UK

Estimation and counterfactuals

Estimation confirms intuition

	US	UK	$\Delta(\text{US-UK})$
Early-stage meeting, ν_e	2.54	1.09	+1.45***
Late-stage meeting, ν_l	5.20	2.28	+2.92***
Acquisition offers, ϕ	0.65	0.33	+0.32***

- Other parameters not statistically distinguishable.

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- **Financing.** Impose UK financing conditions (ν_e, ν_l) on US projects.
 - Financing explains **two-thirds** of the creation-incentive gap, $V_{s,e}$.

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Counterfactuals

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 - Financing explains **two-thirds** of the creation-incentive gap, $V_{s,e}$.
- **Policy.** Reallocate 5% of UK capital from early- to late-stage firms, holding investment fixed.
 - More firms reach success, with higher payoffs.
 - Total surplus rises by **16–30%**.

Today

- A tractable equilibrium model of **matching and staged financing**.
- Funding histories distinguish **project characteristics** from **financing conditions**.
- Applications to **premature failure in the US** and **cross-country differences in the UK**.

Also in the paper

- **Theory.** Equilibrium contracting, externalities and efficiency.
- **Quantitative.** Entry distortions and capital misallocation.

Appendix

Appendix map

Model and identification

- . Quantitative extensions
- . Staging and syndication
- . Contracting microfoundations
- . Alternative contract
- . Closing model
- . Funding-history comparative statics
- . Equilibrium results
- . Social-planner benchmark
- . Identification formulas

Data and United States

- . Illustrative dataset
- . Sample overview
- . Fit details
- . Parameter estimates
- . Failure decomposition
- . Capital misallocation

Appendix map: robustness and United Kingdom

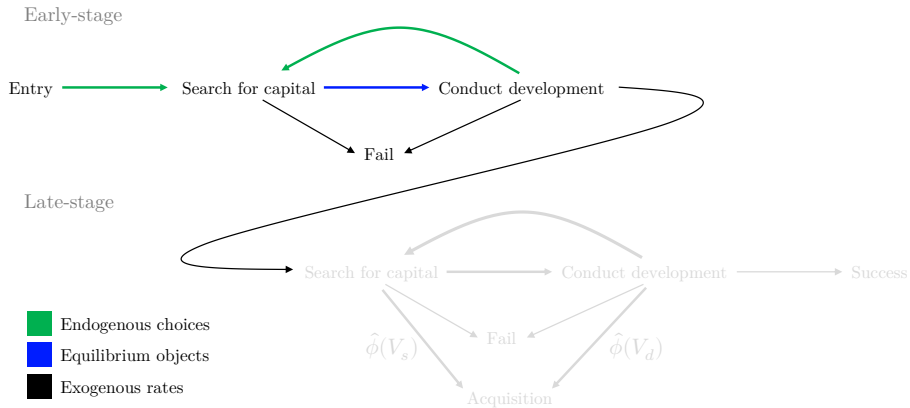
Robustness and auxiliary evidence

- . Identifying acquisitions
- . Censoring and sample period
- . Burn-rate sensitivity
- . US historical evidence (ERISA)
- . US-UK tax treatment
- . Market segmentation
- . Cash flow at IPO

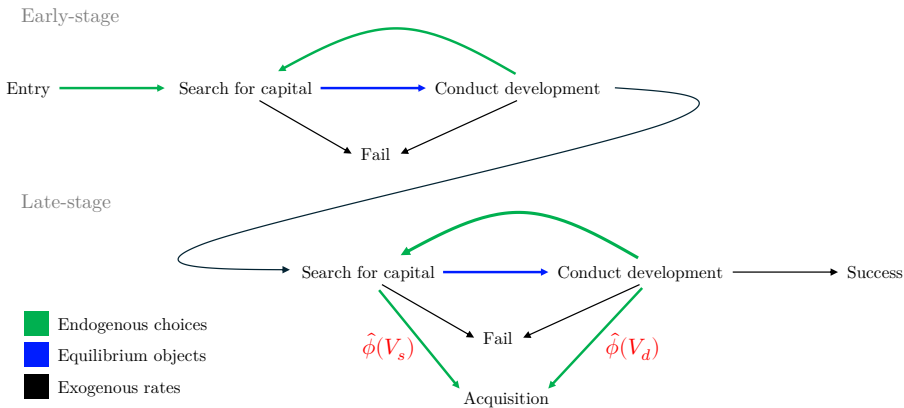
United Kingdom

- . Key estimates and counterfactual
- . Targeted and untargeted moments
- . Full parameter comparison
- . UK model fit
- . US-UK comparison
- . Alternative explanations
- . Policy counterfactual

Quantitative extensions



Quantitative extensions



- **Acquisitions substitute for financing:** $\hat{\phi}(V_s) > \hat{\phi}(V_d)$. [Details](#)

Acquisitions substitute for financing

- Introducing acquisitions
 - Late-stage start-ups meet potential acquirors at rate ϕ .
 - Acquirors value the firm at $\varepsilon p_l \pi$, where $\varepsilon \sim \text{Exp}(1/\xi)$ is *iid* and common knowledge.
 - The start-up makes a take-it-or-leave-it offer:

$$\text{accept if } \varepsilon p_l \pi \geq V_{i,l}, \quad i \in \{s, d\}.$$

- The endogenous acquisition arrival rate is

$$\hat{\phi}_i = \phi \left[1 - F\left(\frac{V_{i,l}}{p_l \pi}\right) \right], \quad i \in \{s, d\}.$$

- **Key result.** The acquisition rate is higher in search than development: $\hat{\phi}_s > \hat{\phi}_d$.

Illustrative dataset

Illustrative dataset					
Firm ID	Date	Event	Round Type	Amount raised	Valuation
1	28/01/2013	funding round	Early	6.0	-
1	24/09/2013	funding round	Early	6.5	-
1	04/02/2015	funding round	Late	30.0	-
1	17/04/2019	IPO	-	80.5	342.6
2	12/03/2013	funding round	Early	3.0	-
2	19/07/2016	funding round	Late	10.5	-
2	01/01/2017	Acquisition	-	-	89.4
3	06/03/2014	funding round	Early	9.1	-

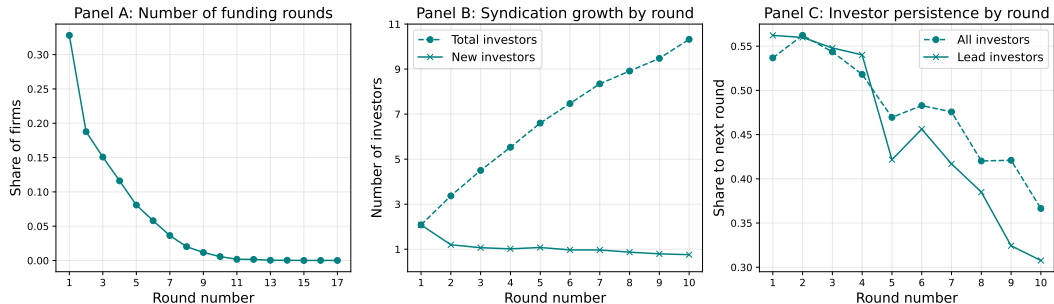
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Sample overview

	United States		United Kingdom	
	Count	Share	Count	Share
<i>Sample overview</i>				
Firms	11,030	–	1,066	–
Successes	400	3.6%	26	2.4%
Acquisitions	2,034	18.4%	97	9.1%
Funding rounds—total	33,158	–	2,218	–
Early-stage	21,886	66.0%	1,534	69.2%
Late-stage	11,272	34.0%	684	30.8%
<i>Sectors</i>				
Biotechnology	995	9.0%	97	9.1%
Communications and Media	289	2.6%	43	4.0%
Computer Hardware	298	2.7%	31	2.9%
Computer Software and Services	3,607	32.7%	254	23.8%
Consumer Related	328	3.0%	53	5.0%
Industrial/Energy	537	4.9%	99	9.3%
Internet Specific	3,082	28.0%	246	23.1%
Medical/Health	1,063	9.6%	109	10.2%
Other Products	463	4.2%	85	8.0%
Semiconductors/Other Elect.	368	3.3%	49	4.6%
Total	11,030	100.0%	1,066	100.0%

[Back to quantitative model](#)[Back to alternative explanations](#)

Staging and investor syndication



Source: Thomson Reuters, US VC-backed start-ups first funded 2005–2015, data to end-2022. Censored at 7 years (panel A) and 12 years (panel B).

[Back to model](#)

Alternative contracts [1/2]

- In practice, a VC invests K for equity share ς , implying runway $T_K = K/k$.
 - Uncertainty in the burn rate k makes runway uncertain.
- **Alternative setting.**
 - The contract takes the form (K, ς) .
 - The entrepreneur draws run-rate $\tilde{k}$ after investment, with $\mathbb{E}[\tilde{k}] = k$.
 - If the project succeeds, unused capital is returned to the VC.
- Expected capital cost

$$\tilde{K}(K) = K \left(1 - \frac{\rho \kappa}{\rho + \kappa} \right) + \frac{\rho \kappa}{(\rho + \kappa)^2} \mathbb{E}_{\tilde{k}} \left[\tilde{k} \left(1 - e^{-(\rho + \kappa)K/\tilde{k}} \right) \right].$$

→ Qualitatively similar to the ω -contract.

Alternative contracts [2/2]

- Example: $1/\tilde{k} \sim \Gamma(\text{shape} = 2, \text{rate} = k)$.

$$\tilde{K}(K) \Big|_{\rho \rightarrow 0} = \frac{k}{\kappa + k/K} \left(1 + \frac{\kappa(1-p)}{k/K} \right), \quad \mathbb{E}[1/T_K] = k/K.$$

- Compare with

$$K(\omega) \Big|_{\rho \rightarrow 0} = \frac{k}{\kappa + \omega} \left(1 + \frac{\kappa(1-p)}{\omega} \right), \quad 1/\mathbb{E}[T_\omega] = \omega.$$

[Back to contract](#)

Contracting problem

Set-up

- **Private benefits.** The entrepreneur derives non-pecuniary benefit $x_b < k$ for the duration of the contract.
- **Private information.** The realisation of uncertainty at $T_\kappa \sim \text{Exp}(\kappa)$ is observed by the entrepreneur but not the VC.

[Back to model](#)

Contracting problem

VC contracts

- Equity contracts take the following form:
 - The VC obtains share ς of post-money value V_d .
 - The VC commits to fund development cost k for a certain duration.
- Funding duration is determined jointly by
 - [i] a termination rule at T_κ , which may condition on success or failure;
 - [ii] a Poisson contract rate ω , at which the funding commitment terminates.

Contracting problem

[i] The termination rule

- Termination at T_κ requires the entrepreneur to disclose the private signal.
 - The termination rule must be **incentive compatible**.
- It is never incentive compatible to reveal negative news: the contract **cannot condition on failure**.
 - **Inefficient continuation**.
- Revealing positive news requires sufficient **skin in the game**.
- **Key result**. For any parameters, there is a $\bar{x}_b > 0$ such that, if $x_b < \bar{x}_b$, the entrepreneur reports positive results in the optimal contract but not negative results.

Contracting problem

[ii] The contract rate

- As $x_b \rightarrow 0$, the entrepreneur reports success immediately but engages in inefficient continuation after failure.
- The expected discounted capital cost is

$$K(\omega) = \underbrace{\frac{k}{\rho + \kappa + \omega}}_{\text{productive capital}} + \underbrace{\frac{\kappa(1-p)}{\kappa + \omega}}_{\text{Pr(failure)}} \underbrace{\frac{k(\kappa + \omega)}{(\rho + \omega)(\rho + \kappa + \omega)}}_{\text{capital after failure}} = \left(1 + \frac{\kappa(1-p)}{\rho + \omega}\right) \frac{k}{\rho + \kappa + \omega}.$$

- Agency conflict creates a wedge between internal and external funds:
 - [i] $\omega \uparrow$ lowers the wedge (staged financing);
 - [ii] $p \downarrow$ raises the wedge;
 - [iii] $\kappa \uparrow$ raises the wedge (milestone financing).

Identification: baseline model

$$\mathbb{E}[N_f] = \frac{(\kappa + \omega)(\lambda + \nu)}{\kappa(\lambda + \nu) + \lambda\omega}$$

$$\mathbb{E}[T_{br}] = \frac{1}{\kappa + \omega} + \frac{1}{\lambda + \nu}$$

$$\mathbb{E}[1\{\text{success}\}] = \mathbb{E}[N_f] \frac{\kappa\rho}{\kappa + \omega}$$

$$\mathbb{E}[\text{time-to-exit}] = \frac{(\lambda + \nu)^2 + \nu\omega}{(\lambda + \nu)(\kappa(\lambda + \nu) + \lambda\omega)}$$

$$\mathbb{E}[\text{burn rate}] = K(\omega) \frac{(\kappa + \omega)(\lambda + \nu) \ln\left(\frac{\kappa + \omega}{\lambda + \nu}\right)}{(\kappa + \omega) - (\lambda + \nu)}$$

$$\mathbb{E}[\text{exit multiple}] = \frac{\pi}{K(\omega)} \frac{\mathbb{E}[N_f]^{-1}}{1 - \mathbb{E}[N_f]^{-1}} \ln(\mathbb{E}[N_f])$$

Proposition 2

The **expected number of funding rounds** $\mathbb{E}[N_f]$ is

- increasing in the meeting rate ν and flow investment cost k ;
- decreasing in R&D success p , payoff π , and search failure λ ;
- ambiguous in the arrival rate of R&D outcomes κ .

Holding project quality $\bar{V}_s$ fixed, $\mathbb{E}[N_f]$ is

- decreasing in the R&D success probability p ;
- ambiguous in the arrival rate κ .

Proposition 3

The *expected duration between successive rounds* $\mathbb{E}[T_{br}]$ is

- increasing in R&D success p and payoff π ;
- decreasing in the meeting rate ν and flow investment cost k ;
- ambiguous in search failure λ and outcome arrival κ .

Holding project quality $\bar{V}_s$ fixed, $\mathbb{E}[T_{br}]$ is

- increasing in the R&D success probability p ;
- strictly decreasing in the arrival rate κ .

Assessing model fit: targeted and untargeted moments

Panel A: Targeted moments

Moment	Data	Model
Mean # early rounds	1.98	1.91
Mean # late rounds late	2.46	2.43
Share one round (%)	32.81	26.47
Mean duration b/w rounds	1.16	1.25
Estimate of $\lambda + \nu_e$	3.30	3.16
Estimate of $\lambda + \nu_l + \hat{\phi}_s$	5.94	6.15
Share funded after year 5 (%)	19.68	18.25
Mean burn rate early	7.91	7.49
Mean burn rate late	23.78	24.89
Share to late-stage (%)	41.56	41.86
Mean time-to-exit	3.67	3.33
Acquisition-to-success ratio	5.08	5.09
Mean exit multiple success	9.24	9.06
Share acquisition multiples > 10X	8.01	8.17
# first funding rounds	1.00	1.00

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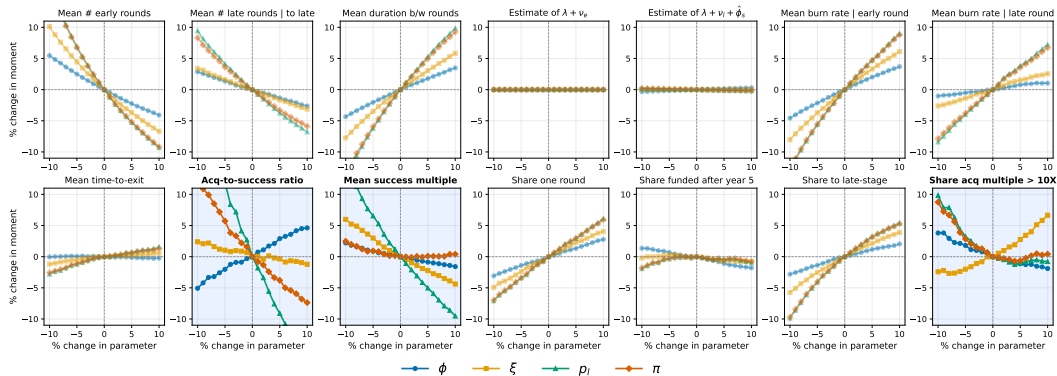
Panel B: Untargeted moments

Moment	Data	Model
Share successful (%)	3.63	3.78
Share acquired (%)	18.44	19.24
Mean time-to-success	4.44	3.40
Mean time-to-acquisition	3.52	3.31
Mean # funding rounds	3.01	2.93
Mean exit multiple	5.79	5.33

US parameter estimates

Parameter		Estimate	Std. Error
Discount rate	ρ	0.08	–
R&D milestone arrival rate	κ	0.26	0.01
Failure rate in search	λ	0.62	0.04
Early-stage development level	p_e	0.23	0.01
Early-stage flow cost (\$Mn/year)	k_e	4.16	0.18
Early-stage meeting rate	ν_e	2.54	0.13
Late-stage development level	p_l	0.30	0.01
Late-stage flow cost (\$Mn/year)	k_l	16.15	0.25
Late-stage meeting rate	ν_l	5.20	0.26
Success payoff (\$Mn)	π	218.18	15.40
Acquisition offer arrival rate	ϕ	0.65	0.07
Mean acquisition synergy	ξ	0.74	0.08
Entry cost distribution	$\tilde{\sigma}$	2.68	0.37

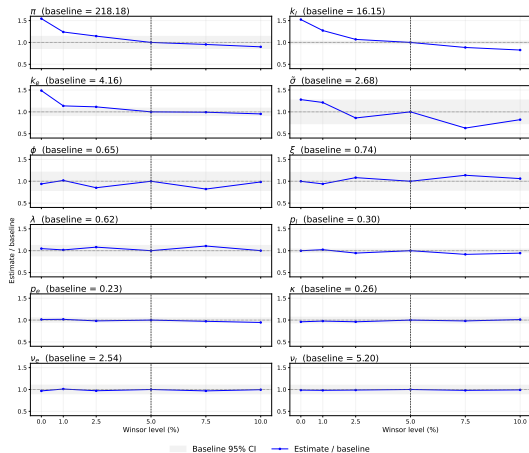
Comparative statics



Parameter estimates under alternative sample periods and censoring

Parameter	US, 2005–2015 (C7)		US, 2005–2010 (C7)		US, 2005–2010 (C12)	
	Estimate	s.e.	Estimate	s.e.	Estimate	s.e.
ρ	0.08	–	0.08	–	0.08	–
κ	0.26	0.01	0.25	0.03	0.26	0.02
λ	0.62	0.04	0.71	0.10	0.64	0.15
ρ_e	0.23	0.01	0.18	0.01	0.18	0.01
k_e	4.16	0.18	3.81	0.35	3.79	0.23
ν_e	2.54	0.13	2.68	0.25	2.75	0.18
ρ_l	0.30	0.01	0.21	0.01	0.24	0.01
k_l	16.15	0.25	11.44	0.66	11.81	1.06
ν_l	5.20	0.26	5.87	0.63	6.70	0.66
π	218.18	15.40	223.42	32.07	220.95	29.39
ϕ	0.65	0.07	0.46	0.12	0.44	0.10
ξ	0.74	0.08	0.98	0.13	1.01	0.20
$\tilde{\sigma}$	2.68	0.37	1.60	1.00	2.80	1.11

Sensitivity to burn-rate winsorisation



UK moments

Moment	Data	Model
Mean # early rounds	1.44	1.44
Mean # late rounds late	1.94	1.69
Share one round	0.53	0.43
Mean duration b/w rounds	1.46	1.72
Estimate of $\lambda + \nu_e$	2.13	1.90
Estimate of $\lambda + \nu_l + \hat{\phi}_s$	2.99	3.31
Share receiving funding after year 5	0.12	0.12
Mean burn rate early	5.80	5.42
Mean burn rate late	18.79	19.50
Share to late-stage	0.33	0.34
Mean time-to-exit	3.82	3.31
Acquisition-to-success ratio	3.73	3.63
Mean exit multiple success	8.88	8.97
Share acquisition multiples > 10X	0.11	0.11
# first funding rounds	0.48	0.48
Untargeted		
Share successful	0.02	0.03
Share acquired	0.09	0.13

[Back to UK estimates](#)

Full US–UK parameter comparison

Parameter	US		UK		Difference US–UK
	Estimate	s.e.	Estimate	s.e.	
κ	0.26	0.01	0.28	0.06	-0.02
λ	0.62	0.04	0.81	0.17	-0.19
ρ_e	0.23	0.01	0.26	0.07	-0.03
k_e	4.16	0.18	3.46	1.02	0.70
ν_e	2.54	0.13	1.09	0.19	1.45***
ρ_l	0.30	0.01	0.30	0.04	0.00
k_l	16.15	0.25	12.04	2.53	4.11
ν_l	5.20	0.26	2.28	0.41	2.92***
π	218.18	15.40	205.26	66.56	12.92
ϕ	0.65	0.07	0.33	0.07	0.32***
ξ	0.74	0.08	0.81	0.72	-0.07
$\tilde{\sigma}$	2.68	0.37	0.98	2.32	1.70

[Back to UK estimates](#)

Alternative explanations

- US institutions/VC governance reduce agency costs
 - Would imply less need for staging, but is inconsistent with observed funding profiles.

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- Differences in required rates of return
 - Results are robust to allowing $\rho_{UK} > \rho_{US}$.

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- **US institutions/VC governance reduce agency costs**
 - Would imply less need for staging, but is inconsistent with observed funding profiles.
- **Differences in required rates of return**
 - Results are robust to allowing $\rho_{UK} > \rho_{US}$.
- **VC screening ability**
 - Would load onto (k, κ, p, π) , but these are similar across markets conditional on funding.

Venture capital and tax regimes: US and UK

1. Corporation tax

- US: 21% plus state-level; UK: 19% (both 2022). **Incorporated in market prices.**

2. Tax on founders

- **Capital-gains rates are similar** across countries ($\sim 20\%$ in 2022), before exemptions.
- Other complications include the treatment of stock options.

3. Tax on investors VC limited partnerships have “look-through” treatment

- **GPs:** carried interest is generally treated as capital gains; management fees as income.
- **LPs:** the effective rate depends on the investor mix and applicable exemptions.
→ UK carry taxation is slightly higher; the LP wedge is ambiguous.

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- **GPs:** carried interest is generally treated as capital gains; management fees as income.
 - **LPs:** the effective rate depends on the investor mix and applicable exemptions.
→ UK carry taxation is slightly higher; the LP wedge is ambiguous.
- **Thought experiment.** Raising the UK required return ($\rho^{UK} = 0.09$ versus $\rho^{US} = 0.08$) leaves the estimated VC-market parameters essentially unchanged.

Market segmentation

- **2002–2020 fund vintages:** median **IRR UK 10.3% versus US 11.8%** (British Business Bank).
- **Who are the LPs in UK-based VC funds?** Approximately one-third UK, one-third rest of Europe, and one-third rest of world (BVCA, 2020).
- **Which VCs invest in UK start-ups?** Approximately one-third of UK VC comes from US investors; the share rises at later stages and has grown over time.
 - US firms with London footprints include Accel, a16z, Bessemer, Lightspeed, General Catalyst, and Sequoia.

Closing the model

- **Value of search.** $(\rho + \lambda)V_s = \nu[V^M - V_s]$, where $V^M = \arg \sup_{\omega} \{(1 - \varsigma)V_d\}$.
- **VCs.** Fixed mass M , can fund one start-up at a time.
- **Entry.** One-shot entry: entrants begin in search with **identical projects** with value V_s .
 - Entry rate $f(V_s)$ with $f'(\cdot) > 0$. [Details](#)
- **Steady-state conditions and matching function.** [Details](#)
- **Equilibrium.** In equilibrium, policy and value functions are consistent with optimality, HJBs, and steady-state conditions.

Start-up entry

- A flow Γ of start-ups have a one-time chance to enter, drawing entry cost $c \sim U[0, \sigma]$.
- Entry is a cutoff rule: enter iff $c \leq \hat{c} = V_s$.
- Given $c \sim U[0, \sigma]$, $\Pr(c \leq V_s) = V_s/\sigma$.
- Γ and σ are observationally equivalent. Define $\tilde{\sigma} = \sigma/\Gamma$.
- The flow rate of entry is $V_s/\tilde{\sigma}$.

Equilibrium conditions

- Steady-state conditions

$$\text{Search: } V_s/\tilde{\sigma} + \omega\mu_d^p = (\lambda + \nu)\mu_s$$

$$\text{Productive development: } \nu\mu_s = (\kappa + \omega)\mu_d^p$$

$$\text{Unproductive development: } \kappa(1 - p)\mu_d^p = \omega\mu_d^u$$

- Meeting rate:

$$\nu = \mu_s^{-1} m(\mu_s, M - \mu_d^p - \mu_d^u) .$$

General Nash bargaining [1/3]

- VC value functions

$$\rho U_d^p = -k + \kappa[p(\varsigma\pi + U_s) + (1-p)U_d^u - U_d^p] + \omega[\varsigma V_s + U_s - U_d^p],$$

$$\rho U_d^u = -k + \omega[U_s - U_d^u],$$

$$\rho U_s = v[U_d^p - U_s],$$

where $v = \mu_{VC}^{-1} m(\mu_s, \mu_{VC})$.

- Contract (ς, ω) solves

$$\max_{\varsigma, \omega} \{ ((1-\varsigma)V_d - V_s)^\delta (U_d^p - U_s)^{1-\delta} \},$$

where $\delta \in [0, 1]$ is existing shareholders' bargaining power.

General Nash bargaining [2/3]

- The optimal contract rate satisfies

$$V'_d(\omega) = -(k + \rho U_s) \left(\frac{1-p}{(\rho + \omega)^2} + \frac{p}{(\rho + \kappa + \omega)^2} \right),$$

where ρU_s is the VC's opportunity cost of investment.

- The optimal contract rate becomes

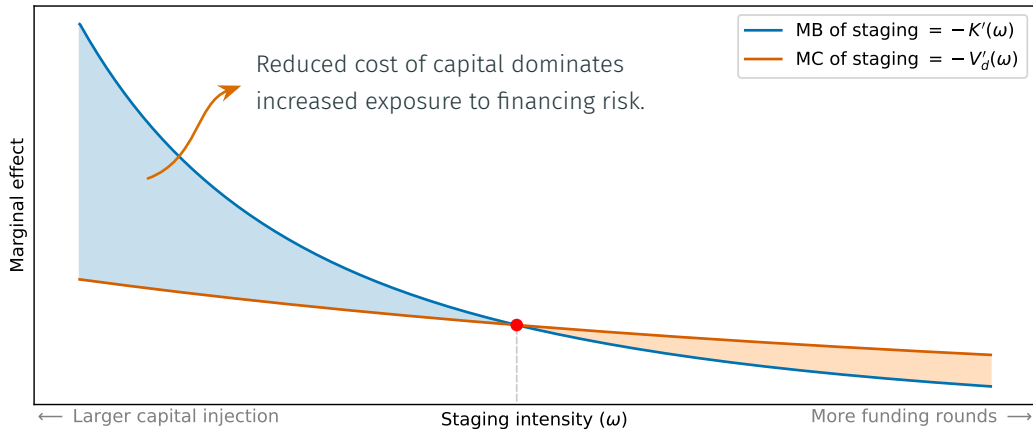
$$\omega = \max \left\{ 0, -\rho + \mathcal{C} + \sqrt{\mathcal{C}\mathcal{F} + \mathcal{C}^2} \right\},$$

$$\mathcal{C} = \frac{\kappa(1-p)k}{\kappa p\pi - k}, \quad \mathcal{F} = \kappa + \delta \nu \frac{\rho + \kappa}{\rho + \lambda} + (1-\delta)v \frac{\rho \kappa \pi}{k}.$$

General Nash bargaining [3/3]

- Bargaining power has two effects
 1. **Preventing future dilution** pushes towards longer runway ($\omega \downarrow$).
 2. The VC's **opportunity cost** pushes towards shorter runway ($\omega \uparrow$).
- $\delta \uparrow$ redistributes returns from VCs to entrepreneurs: $V_s \uparrow$ and $U_s \downarrow$.
 - $V_s \uparrow$: less dilution.
 - $U_s \downarrow$: lower opportunity cost.
- **Ambiguous effect** on the contract rate, ω .

The optimal contract



Equilibrium characterisation

Key model objects

- Funding histories.
- Start-up outcomes.
- Start-up activity.
- Capital misallocation.

[Back to funding histories](#)

Equilibrium characterisation

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- **Funding histories.** A structural interpretation of the number and timing of rounds.
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[Back to funding histories](#)

Equilibrium characterisation

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[Back to funding histories](#)

Equilibrium characterisation

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[Back to funding histories](#)

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- **Capital misallocation.** Premature termination (**underinvestment**) coexists with inefficient continuation (**overinvestment**).

[Back to funding histories](#)

Equilibrium characterisation

The optimal contract

Proposition 4

Denote expected runway by $\tau = 1/\omega$. Then

- *Weaker funding* ($\nu \downarrow$ or $\lambda \uparrow$) *increases runway*, $\tau \uparrow$.
- *Higher surplus* ($p \uparrow, \pi \uparrow, k \downarrow$) *increases runway*, $\tau \uparrow$.
- *Faster completion* ($\kappa \uparrow$) *has an ambiguous effect on runway*.

Equilibrium characterisation

The optimal contract

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Holding project quality $\bar{V}_s = (\kappa p \pi - k)/(\rho + \kappa)$ constant

- More “novel” projects ($p \downarrow$) obtain **less runway**, $\tau \downarrow$.
- Long-horizon projects ($\kappa \downarrow$) obtain **more runway**, $\tau \uparrow$.

Equilibrium characterisation

Start-up outcomes

- Among funded firms, $p_d = (1 - \Delta_p)p \leq p$.

Equilibrium characterisation

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Lemma 1

Consider start-ups a and b facing the same funding conditions. If $\mathbb{E}[N_f^a] > \mathbb{E}[N_f^b]$, then $\Delta_p^a > \Delta_p^b$.

Equilibrium characterisation

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Lemma 1

Consider start-ups a and b facing the same funding conditions. If $\mathbb{E}[N_f^a] > \mathbb{E}[N_f^b]$, then $\Delta_p^a > \Delta_p^b$.

- Higher surplus ($p \uparrow, \pi \uparrow, k \downarrow$) reduces the outcome distortion, Δ_p .
- Holding project quality fixed, the distortion is
 - larger for more novel projects ($p \downarrow, \pi \uparrow$);
 - typically larger for long-horizon projects ($\kappa \downarrow, \pi \uparrow$).

Equilibrium characterisation

Start-up activity

Start-up creation is tied to V_s :

$$V_s|_{\rho \rightarrow 0} = \frac{\nu}{\lambda + \nu} (1 - \Delta_\rho) \frac{\kappa p \pi - k \left(1 + \frac{\kappa(1-\rho)}{\omega}\right)}{\kappa} \leq \frac{\kappa p \pi - k}{\kappa} = \bar{V}_s|_{\rho \rightarrow 0}.$$

- Some firms **never receive funding**, while others **close prematurely**.
- External finance **raises the effective capital cost**.

Equilibrium characterisation

Start-up activity

Start-up creation is tied to V_s :

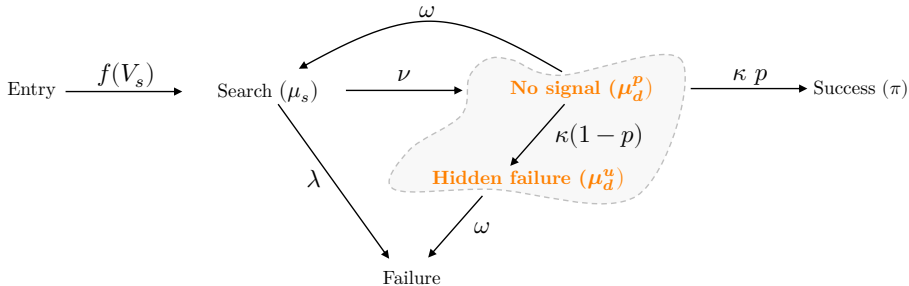
$$V_s|_{\rho \rightarrow 0} = \frac{\nu}{\lambda + \nu} (1 - \Delta_p) \frac{\kappa p \pi - k \left(1 + \frac{\kappa(1-p)}{\omega}\right)}{\kappa} \leq \frac{\kappa p \pi - k}{\kappa} = \bar{V}_s|_{\rho \rightarrow 0}.$$

Proposition 5

- Weaker funding decreases entry value, V_s .
- Higher surplus and faster completion increase entry value, V_s .

Equilibrium characterisation

Capital misallocation



Misallocation $:= \frac{\mu_d^u}{\mu_d^u + \mu_d^p}$ (share of funded firms in hidden failure state).

- Tight markets ($\nu \downarrow$) $\rightarrow$ longer runway $\rightarrow$ misallocation $\uparrow$.

Capital misallocation: comparative statics

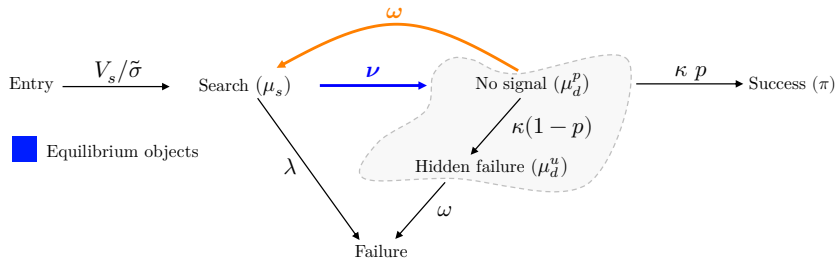
$$\underbrace{\kappa(1-p)\mu_d^p = \omega\mu_d^u}_{\text{steady state}} \implies \frac{\mu_d^u}{\mu_d^p + \mu_d^u} = \frac{\kappa(1-p)}{\kappa(1-p) + \omega}.$$

Proposition 6

Capital misallocation is

- *increasing in the payoff π and the failure rate in search λ ;*
- *decreasing in the meeting rate ν and flow investment cost k ;*
- *ambiguous in R&D success p and the arrival rate of R&D outcomes κ .*

Planner problem [1/2]



Start-ups compete for capital **across cohorts**.

They do not internalise ($\omega \rightarrow \nu$).

- **Constrained efficiency.** Feedback between frictions

1. Market tightness affects contracts ($\nu \rightarrow \omega$).
2. Contracts affect market tightness ($\omega \rightarrow \nu$).

Back to contract

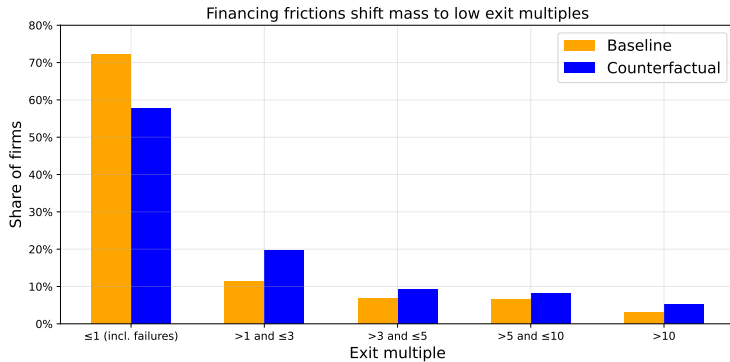
- **Hosios condition.**

Planner problem [2/2]

$$TS = \int_0^\infty e^{-\rho t} \left([\kappa p \pi - k] \mu_{d,t}^p - k \mu_{d,t}^u - \frac{\tilde{\sigma}}{2} \Lambda_t^2 \right) dt.$$

- The planner chooses contracts, ω_t , and entry, Δ_t , subject to the state-transition equations.
- The planner **internalises the effect** of contracts and entry on meeting rates, ν_t and v_t .
- The decentralised equilibrium need not implement this allocation: contract and entry choices create a **matching externality**.

Exit multiple distribution



Notes: Uncensored simulated outcomes; failures counted in ≤ 1 .

A third takeaway

Capital is misallocated in equilibrium

- **Misallocation.** Premature termination (**underinvestment**) coexists with inefficient continuation (**overinvestment**).
 - **23% of viable firms have no funding** and remain in search.
 - **12% of invested capital** is tied up in firms that have privately learned that the project failed.
- **Implication.** Even holding aggregate capital fixed, its allocation across firms is quantitatively important.

Where does 40% come from?

Decomposition and sensitivity [1/2]

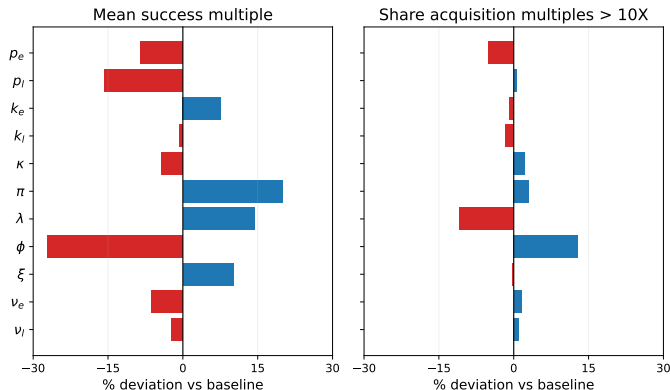
- Early-stage decomposition

$$\Delta_p = \overbrace{\underbrace{\mathbb{E}[N_f]}_{\mathbb{E}[\# \text{ rounds}]} \times \underbrace{\frac{\omega}{\kappa + \omega}}_{\Pr(\text{enter search})}}^{\mathbb{E}[\# \text{ searches}]} \times \underbrace{\frac{\lambda}{\lambda + \nu}}_{\Pr(\text{fail}|\text{search})} = 2.04 \times 0.64 \times 0.20 = 0.26.$$

- The data must pin down
 1. the split of time between development and search; and
 2. contract and search-failure hazards (ω, λ) .
- **Full-model sensitivity:** perturb each moment and re-estimate.

Where does 40% come from?

Decomposition and sensitivity [2/2]



Sensitivity of parameter estimates to a 10% increase in moments, one at a time.

US outcome and distortion breakdown

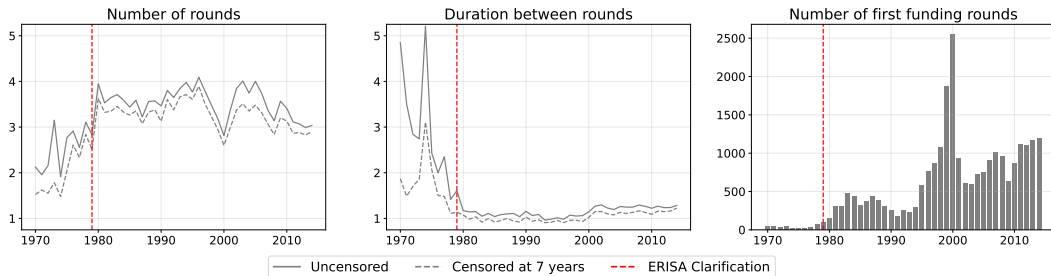
Panel A: Outcomes and modes of failure

Share successful (%)	4.9
Share acquired (%)	24.5
Share failed (%)	70.6
...in early-stage development (%)	18.2
...in early-stage search (%)	25.7
...in late-stage development (%)	11.6
...in late-stage search (%)	15.1

Panel B: Distortions

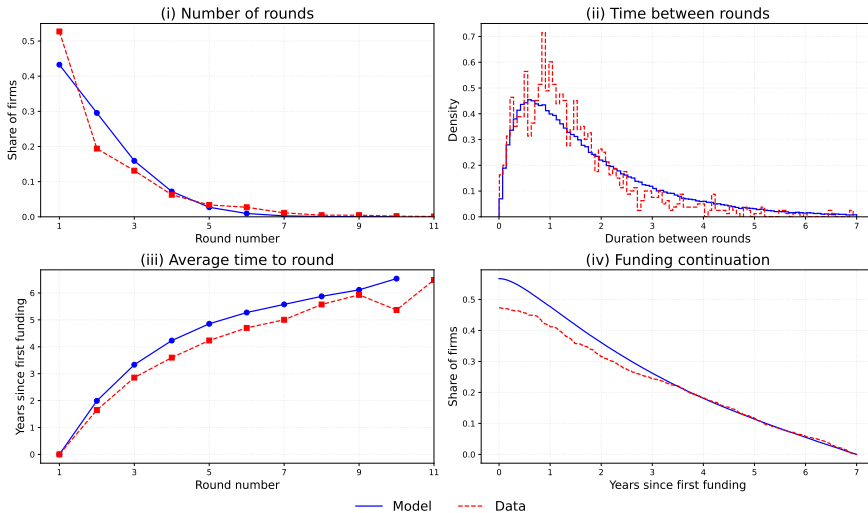
Success distortion, Δ_p (%)	55.5
Acquisition distortion, Δ_a (%)	37.0
Entry distortion, Δ_{NPV} (%)	71.5

ERISA and US funding patterns

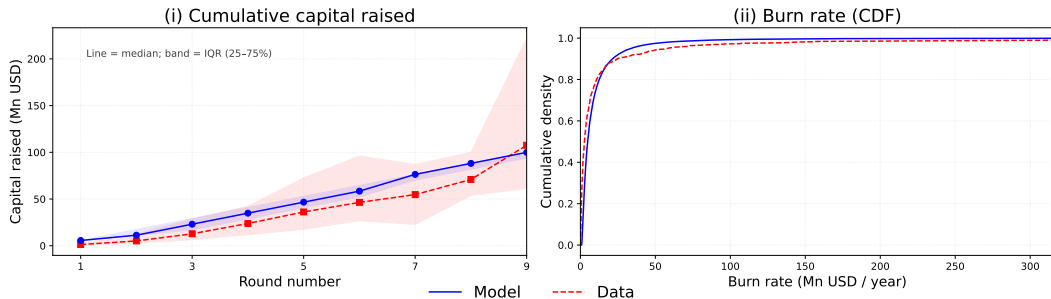


- ERISA led to a large expansion in VC (Gompers & Lerner, 1999).
- The expansion coincides with more rounds and shorter duration between rounds.

UK model fit: funding histories

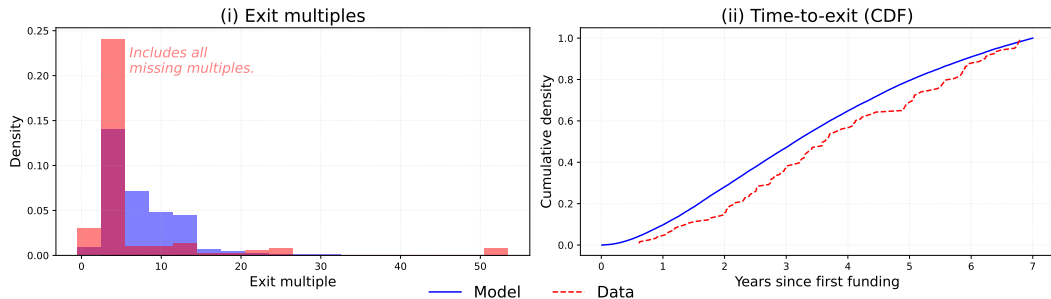


UK model fit: capital intensity



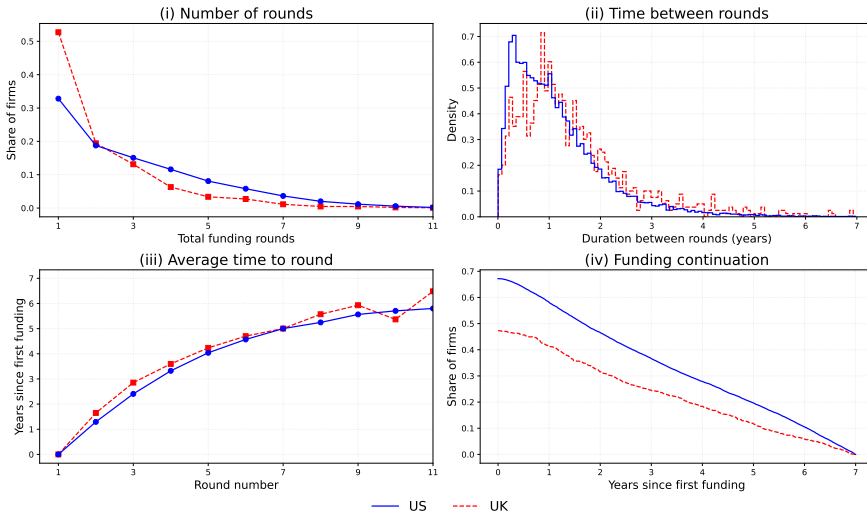
Notes: Cumulative capital by funding round and the burn-rate distribution. Data are censored seven years after first funding.

UK model fit: outcomes



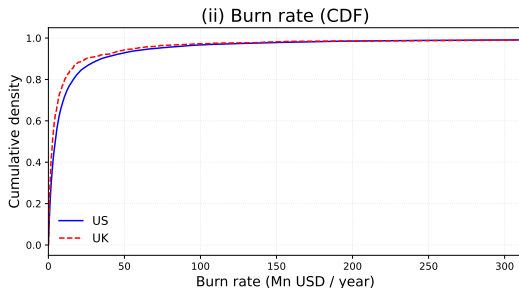
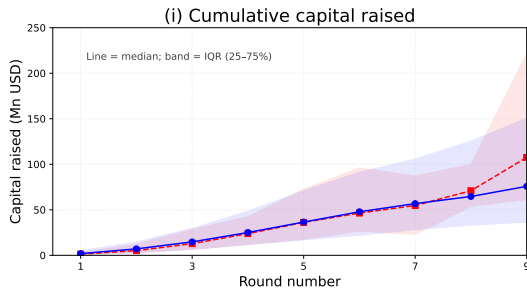
Notes: Exit-multiple distribution conditional on exit and time-to-exit CDF. Data are censored seven years after first funding.

US versus UK: funding histories



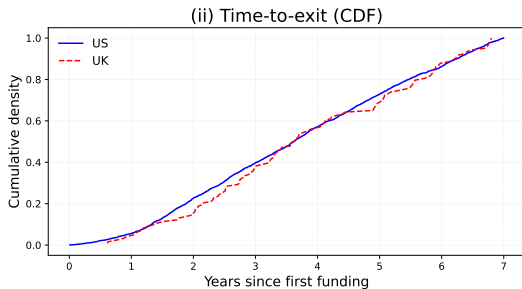
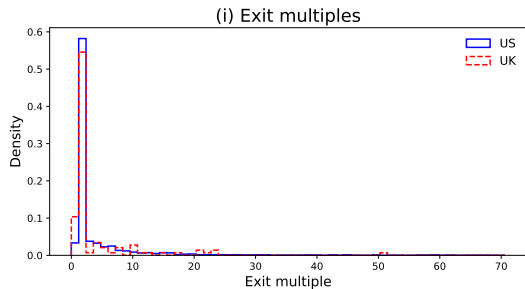
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US versus UK: capital intensity



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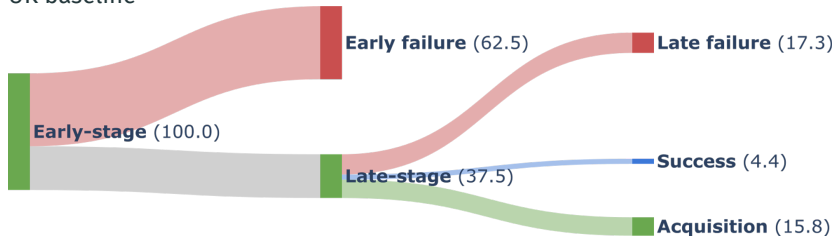
US versus UK: outcomes



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5% capital reallocation: early to late stage

UK baseline

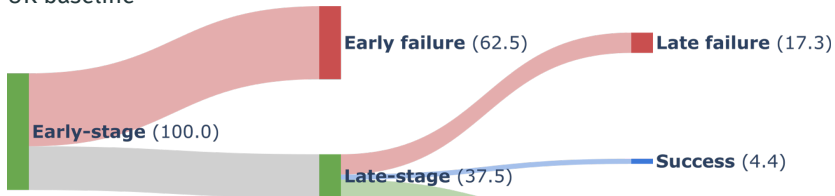


Cash flow at IPO

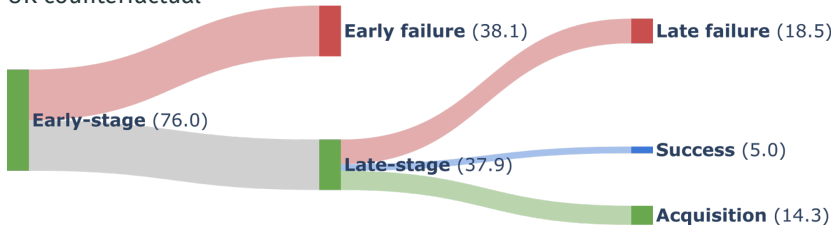
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5% capital reallocation: early to late stage

UK baseline



UK counterfactual

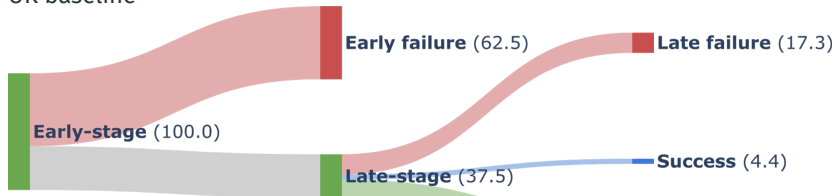


Cash flow at IPO

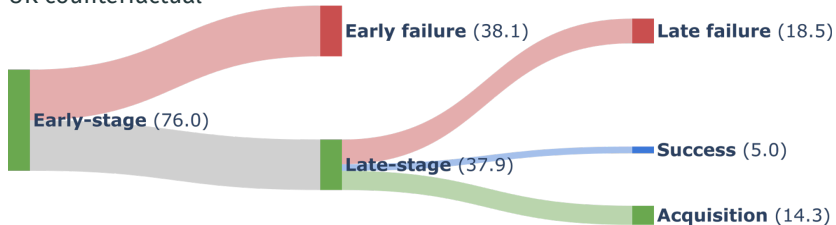
Back to UK estimates

5% capital reallocation: early to late stage

UK baseline



UK counterfactual



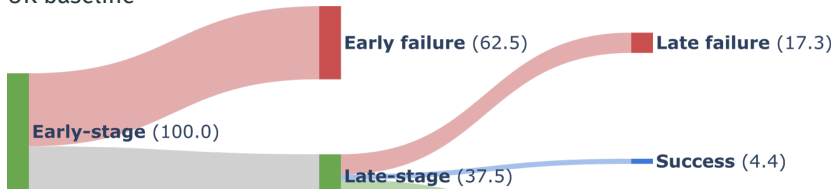
Fewer firms are funded, but more reach success.

Cash flow at IPO

Back to UK estimates

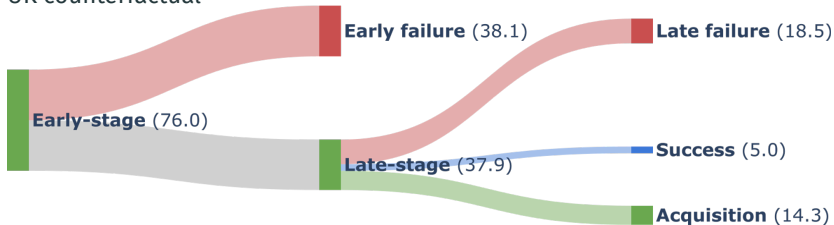
5% capital reallocation: early to late stage

UK baseline



Fewer firms are funded, but more reach success.

UK counterfactual



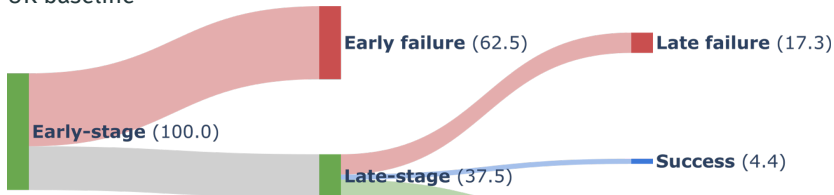
Successful firms have higher payoffs.

Cash flow at IPO

Back to UK estimates

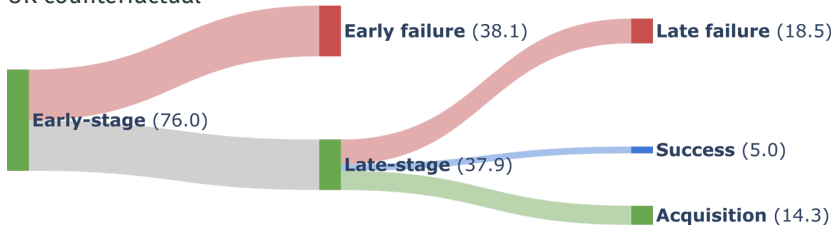
5% capital reallocation: early to late stage

UK baseline



Fewer firms are funded, but more reach success.

UK counterfactual



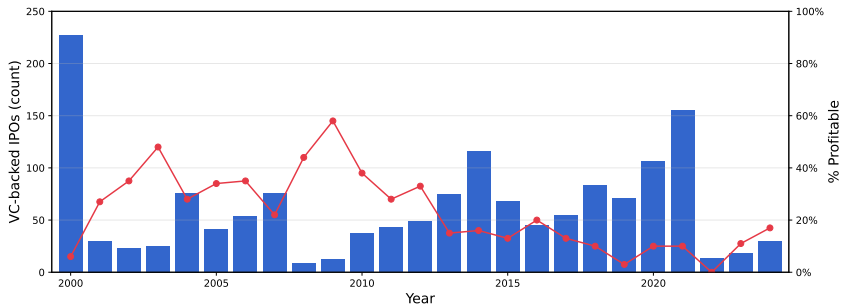
Successful firms have higher payoffs.

Budget-neutral policy: total surplus +16–30%.

Cash flow at IPO

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Start-ups remain cash-flow negative through to IPO



US-headquartered, VC-backed IPOs. Bars report counts; the line reports the share profitable. Source: Jay Ritter, Table 4d.

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